Single-Image Haze Removal via Dark Channel Prior

with a temporally consistent video extension

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One hazy image has too many unknowns; a statistical prior closes the gap

The scattering model

$$I = J \cdot t + A \cdot (1 - t)$$

Seen: I. Wanted: J. Unknown per pixel: A and t. Under-constrained from one image.

Dark Channel Prior (He, Sun, Tang, CVPR 2009): in haze-free outdoor patches, one colour channel is almost always dark.

How good is our implementation?

Full DCP pipeline, validated on synthetic haze and benchmarked on RESIDE SOTS.

What makes video flicker?

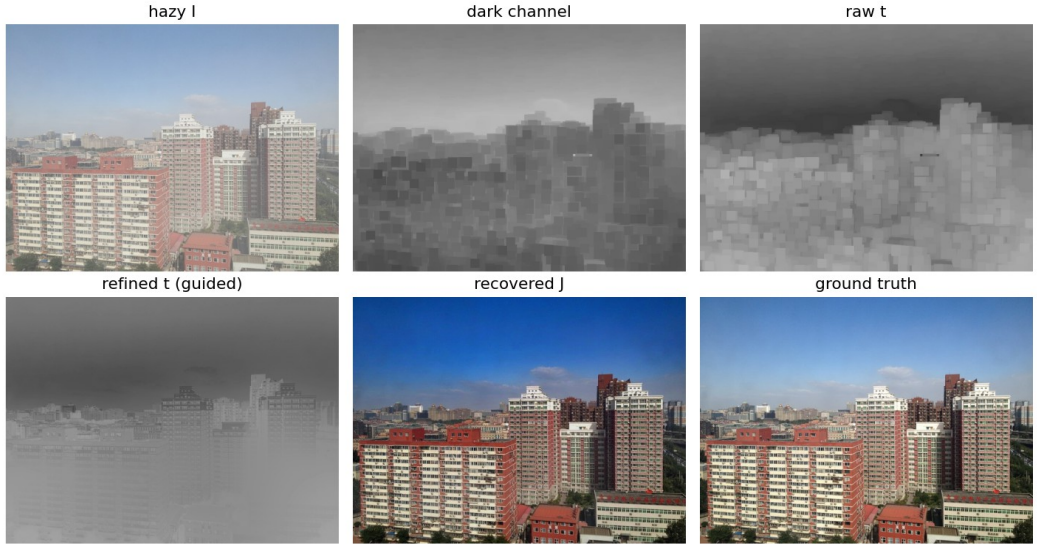
Per-frame DCP jitters because A and t come from extremal statistics.

Can classical smoothing fix it?

Four variants V0–V3, scored against clean ground-truth video (REVIDE).

Five classical stages, every intermediate output exposed

1 Dark channel | 2 Atmospheric light A | 3 Raw transmission t | **4 Refine t** | 5 Recover J



Refiners compared
Soft matting · guided filter · fast guided filter

Synthetic round trip
Known A and t recovered; A within 0.044 per channel

Watch the sky
Recovered J over-saturates it: foreshadows the failure case

Refinement adds 1.3–1.9 dB; soft matting costs ~100× more for no gain

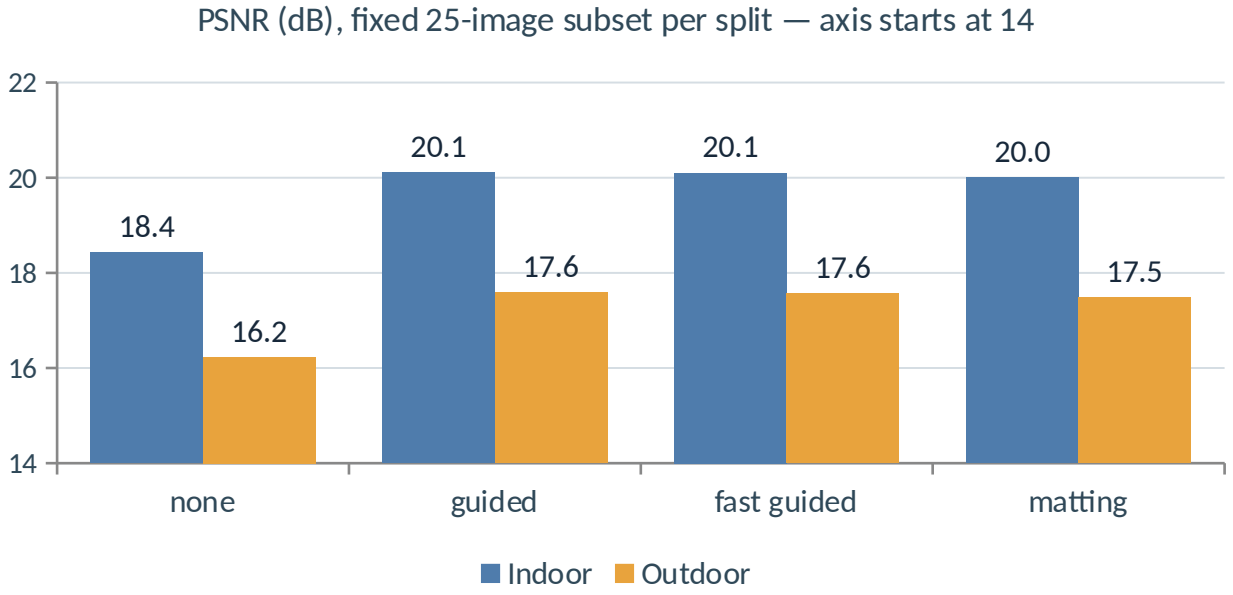
19.82

dB, indoor (SOTS, 500 images)

17.03

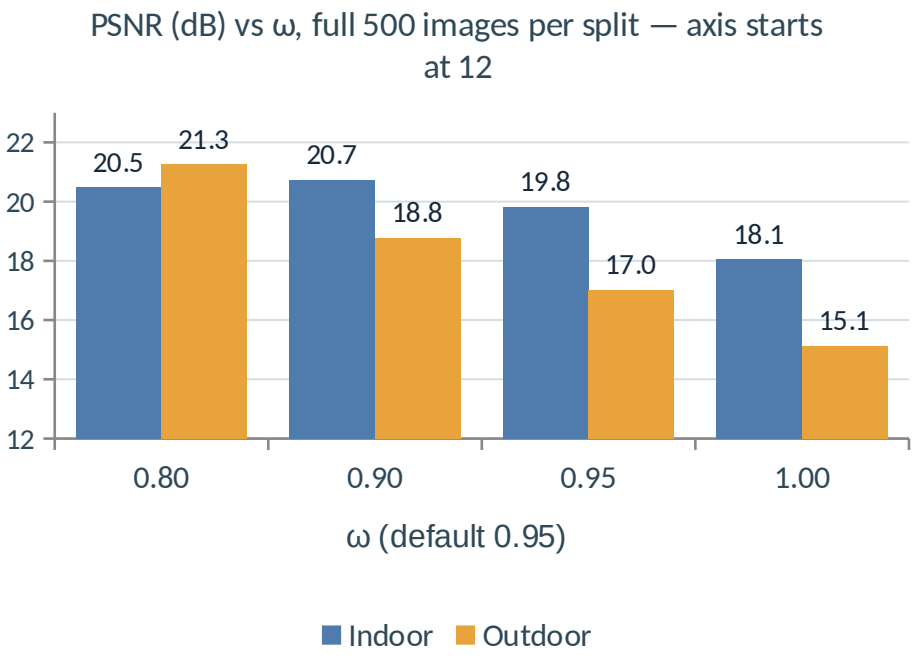
dB, outdoor (SOTS, 500 images)

Defaults: patch 15, ω 0.95, t_0 0.1, guided filter



Seconds per image: none 0.2 · guided 0.3 · fast guided 0.2 · matting ≈ 38

The paper's defaults are not PSNR-optimal on SOTS, and we did not tune on test



Factor	Indoor	Outdoor
patch 7	17.5	15.0
patch 15 (default)	**19.8**	**17.0**
patch 31	21.4	19.2
t_0 0.1 (default)	**19.8**	**17.0**
t_0 0.3	20.3	18.4
A: brightest (default)	**19.8**	**17.0**
A: mean of candidates	19.9	17.4

PSNR in dB, one factor changed at a time. Plausible reason (not verified): SOTS haze is synthetic and uniform, and PSNR rewards conservative dehazing. We keep the paper defaults as the baseline.

Four variants isolate what each kind of temporal smoothing buys

V0

Per-frame DCP

Baseline: nothing shared between frames

V1

EMA on A

Smooth the atmospheric light; re-estimate t under it

V2

V1 + EMA on t

Naive: blend t with the previous frame's t

V3

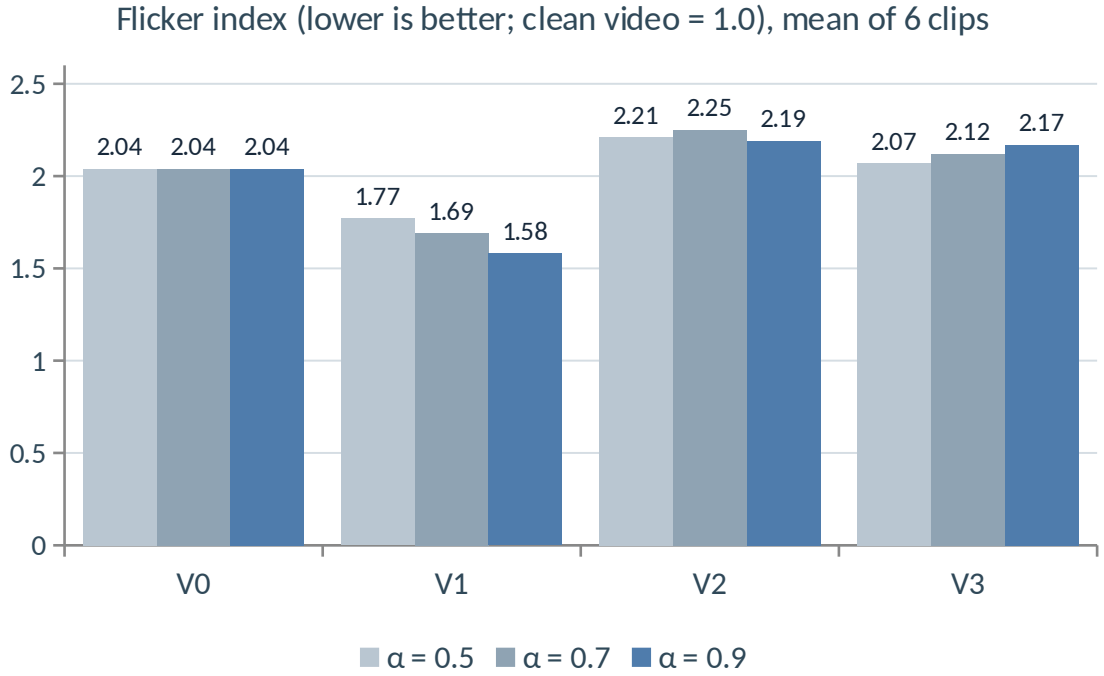
V1 + warped EMA on t

Flow-warp previous t; forward-backward occlusion mask

Flicker index = flow-warped frame difference ÷ the clean video's own (1.0 = natural change; higher = spurious flicker)

REVIDE indoor test split: 6 clips, width 640. Camera moves 5–28 px/frame, so the proposal's static-ROI variance became a grid-block error variance.

Smoothing A cuts flicker; smoothing t does not



2.04 → 1.58

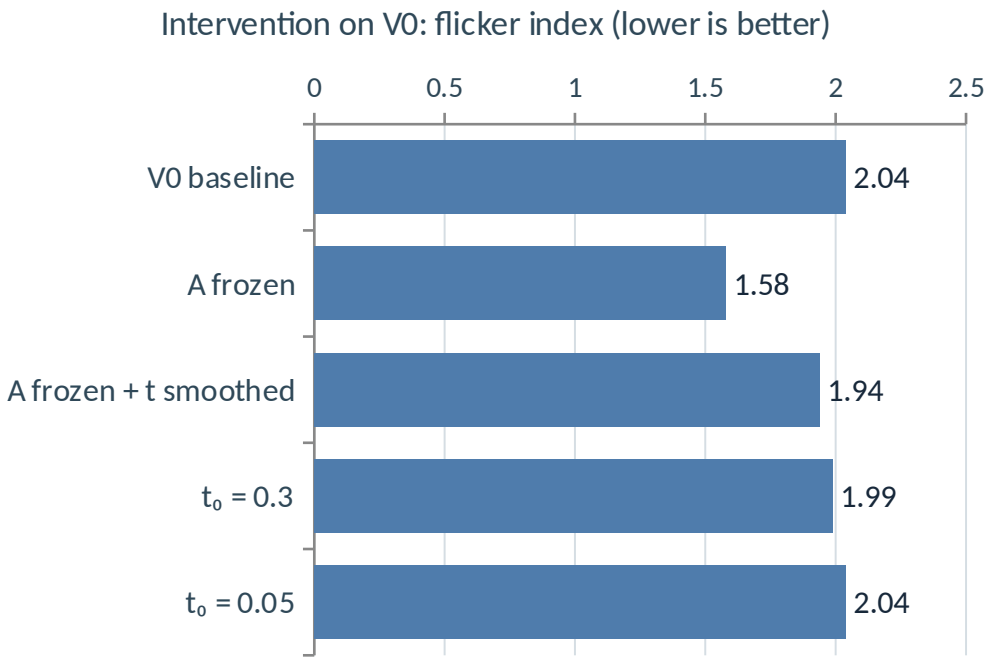
V1 vs V0 at α = 0.9

≈ 2× slower

V3 runs at 5.5 FPS vs 10 for V0 (CPU, width 640)

V3 with ground-truth flow is also worse than V1, so flow quality is not the cause. We found no verified mechanism.

Atmospheric-light jitter is the largest removable cause; 1/t noise gain also shows



Where the flicker lives

≈ 3×

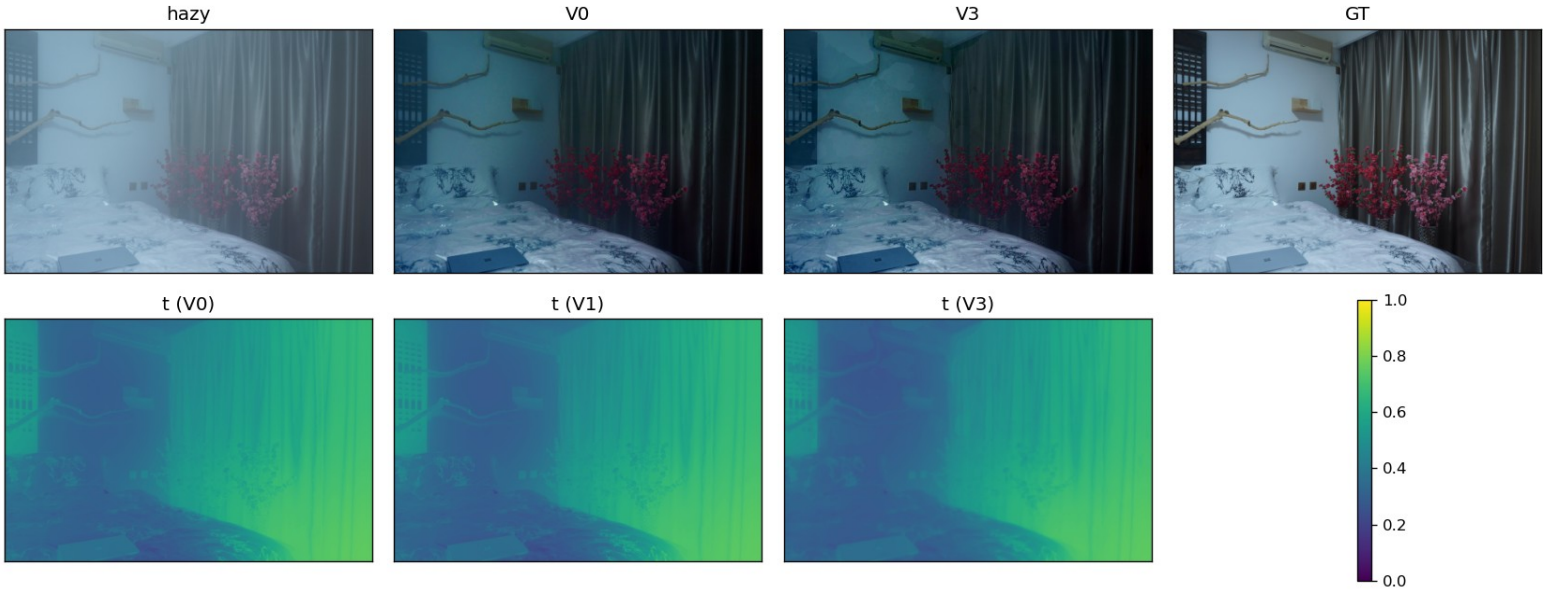
natural flicker in the haziest quartile of t

≈ 1.3×

in the clearest quartile

Consistent with 1/t noise amplification, a fourth cause beyond the proposal's three; not isolated by a direct intervention.

Before and after on clip J004: smoother t, but no less flicker



Full side-by-side clip (hazy | V0 | V1 | V3 | GT): figures/demo_J004.mp4. V3's transmission map is visibly smoother, but that does not lower flicker in the recovered video.

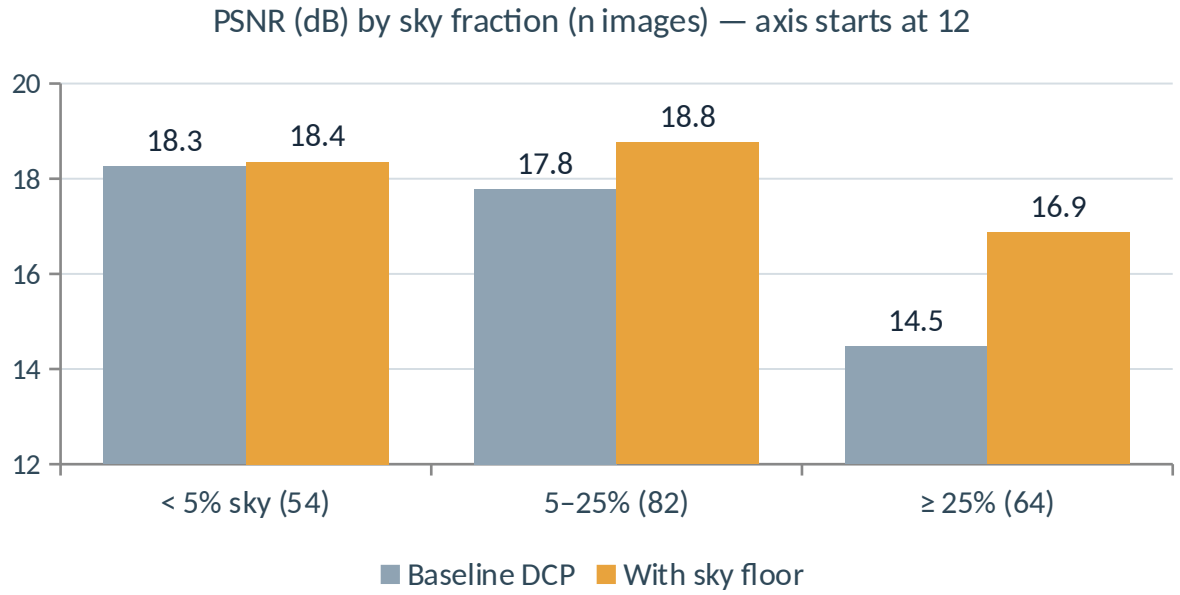
Sky breaks the dark-channel prior; a transmission floor wins back 2.4 dB

r = −0.50

PSNR vs sky fraction, 200 outdoor images

Sky has no dark channel, so t is underestimated and the region is over-amplified.

Mitigation: raise t to 0.6 inside the detected sky region (classical detector, no learning).



Caveat: the floor value and thresholds were chosen once, not tuned, but scored on the same images used for the analysis, so the gain is somewhat optimistic.

What we learned, and what we cannot claim

What worked

Refinement adds 1.3–1.9 dB. Smoothing A cuts flicker (2.04 → 1.58). A sky floor lifts sky-heavy images by 2.4 dB.

What did not

Flow-warped smoothing of t never beat smoothing A alone, even with ground-truth flow. No verified mechanism.

Limits

Indoor-only video, 6 clips, no significance tests, mitigation scored on its analysis set, CPU timings under load.

Everything regenerates with scripts/run_all.sh (verified from empty results/); design decisions and deviations are in IMPLEMENTATION_LOG.md.